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In [ ]: # Task 3: Machine Learning Model – Iris Classification
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In [1]: import pandas as pd
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import accuracy_score, classification_report, cor
```

```
In [2]: df = sns.load_dataset('iris')
df.head()
```

Out[2]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

```
In [3]: df['species'] = df['species'].astype('category').cat.codes
df.head()
```

Out[3]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

```
In [4]: X = df.drop('species', axis=1)
y = df['species']
```

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In [5]: X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

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In [6]: scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

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In [7]: model = LogisticRegression(max_iter=200)
model.fit(X_train, y_train)
```

Out[7]: LogisticRegression(max_iter=200)

```
In [8]: y_pred = model.predict(X_test)
```

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In [9]: accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

Accuracy: 1.0

```
In [10]: print(classification_report(y_test, y_pred))
```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	10
1	1.00	1.00	1.00	9
2	1.00	1.00	1.00	11
accuracy			1.00	30
macro avg	1.00	1.00	1.00	30
weighted avg	1.00	1.00	1.00	30

```
In [11]: print(confusion_matrix(y_test, y_pred))
```

```
[[10  0  0]
 [ 0  9  0]
 [ 0  0 11]]
```

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In [ ]:
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